Economics 2220

Principles Of Macroeconomics

Professor Zhigang Feng

Lecture 13:

Money, Banking and Monetary policy

▶ What you will learn in this chapter

- The various **roles money** plays and the many **forms** it takes in the economy
- How the actions of private banks and the Federal Reserve determine the **money supply**
- How the Federal Reserve uses **open-market operations** to change the **monetary base**

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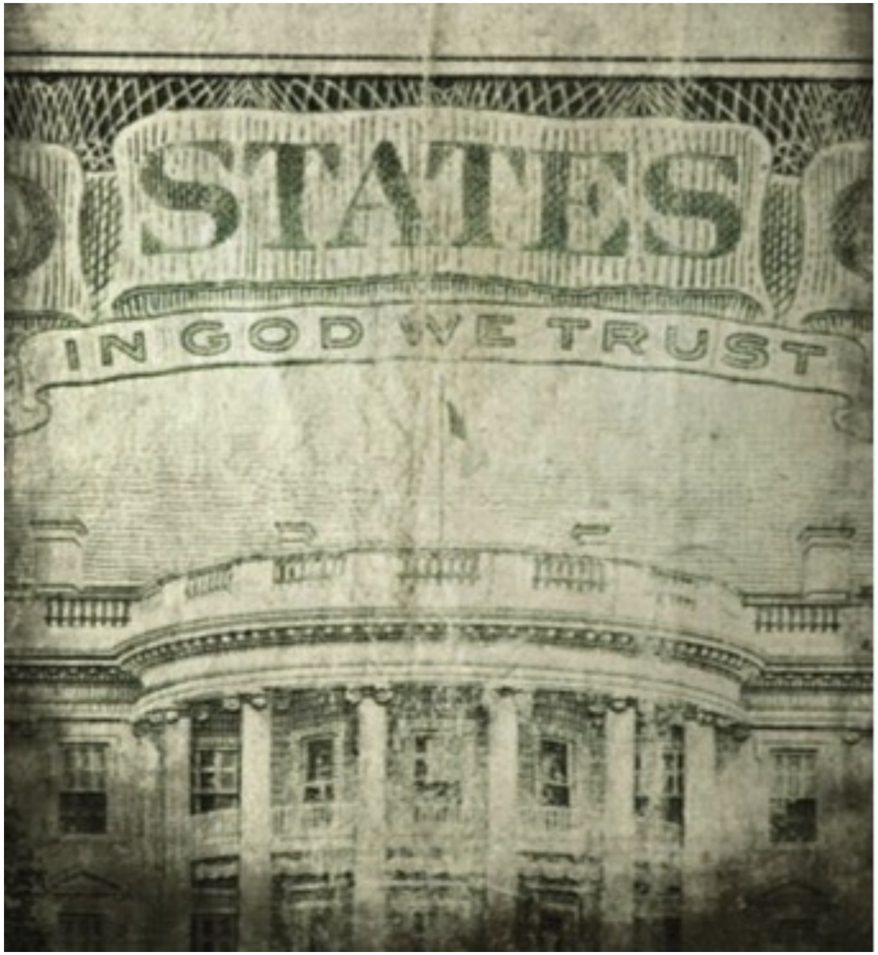
MONEY: THE ESSENTIAL CHANNEL BETWEEN MARKETS

Money plays unique and very important roles in modern economies.



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THE MEANING OF MONEY

Money: **any asset that can easily be used to purchase goods and services.**

In 1933, after many bank failures, some U.S. towns turned to homemade currency, like this clamshell scrip used in Pismo Beach, California.

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THE MEANING OF MONEY

Currency in circulation:
cash held by the public.

Early U.S. currency:
continental
third of a dollar

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THE MEANING OF MONEY

Checkable bank deposits: **bank accounts on which people can write checks.**

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THE MEANING OF MONEY

The *money supply* is the total value of financial assets in the economy that are considered money.

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THE ROLES OF MONEY

Money must function as:

- *a medium of exchange.*
- *a store of value.*
- *a unit of account.*

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THE FUNCTIONS OF MONEY

Medium of exchange: **Something people accept as payment for goods and services**

(an asset that individuals acquire for the purpose of trading rather than for their own consumption).

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THE FUNCTIONS OF MONEY

Store of value: **Money is a means of holding purchasing power over time.**

It enables people to save the money they earn today *and use it to* buy *the goods and services they want* tomorrow.

THE FUNCTIONS OF MONEY

Unit of account: **Money provides a yardstick for measuring and comparing the values of a wide variety of goods and services.**

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TYPES OF MONEY

For thousands of years, societies have used commodity money.

Commodity money: a good used as a medium of exchange that **has intrinsic value in other uses.**

Some prisons use cigarettes as currency.

The island of Yap's currency

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TYPES OF MONEY

To make life simpler (and safer), societies began using commodity-backed money.

Commodity-backed money: a medium of exchange with **no intrinsic value** whose ultimate value is guaranteed by a promise that **it can be converted into valuable goods.**

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ANOTHER EXAMPLE OF COMMODITY-BACKED MONEY

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TYPES OF MONEY

Most modern economies use fiat money.

Fiat money: **money whose value derives entirely from its official status as a means of payment.**

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LEARN BY DOING: DISCUSS

With a partner, discuss:

How would the economy be different if we used only local currency?

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MEASURING THE MONEY SUPPLY

Monetary aggregate: **an overall measure of the money supply.**

M1 and M2: two definitions of the money supply.

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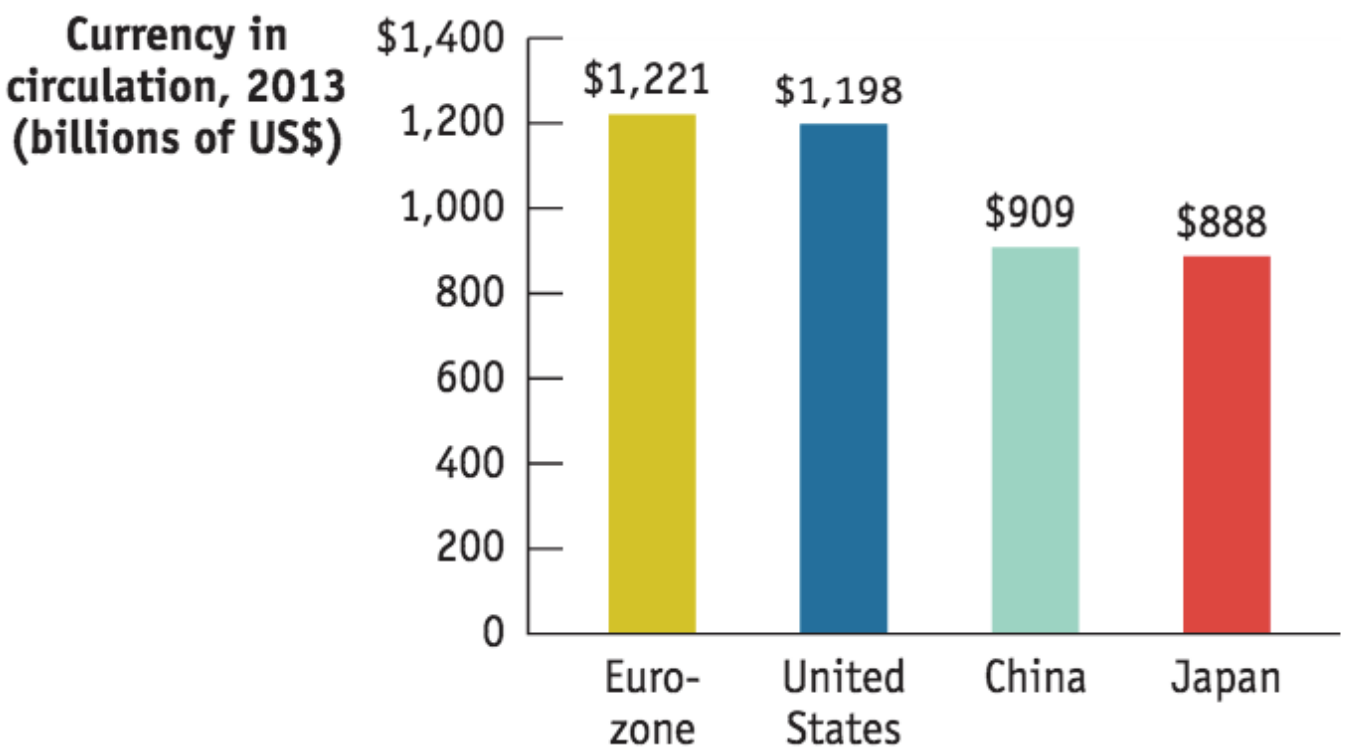
THE BIG MONEYS

The U.S. Dollar is still the currency most likely to be accepted for payment around the world… but is not the biggest money.



Sources: Federal Reserve Bank of St. Louis; European Central Bank; Bank of Japan; The People's Bank of China.

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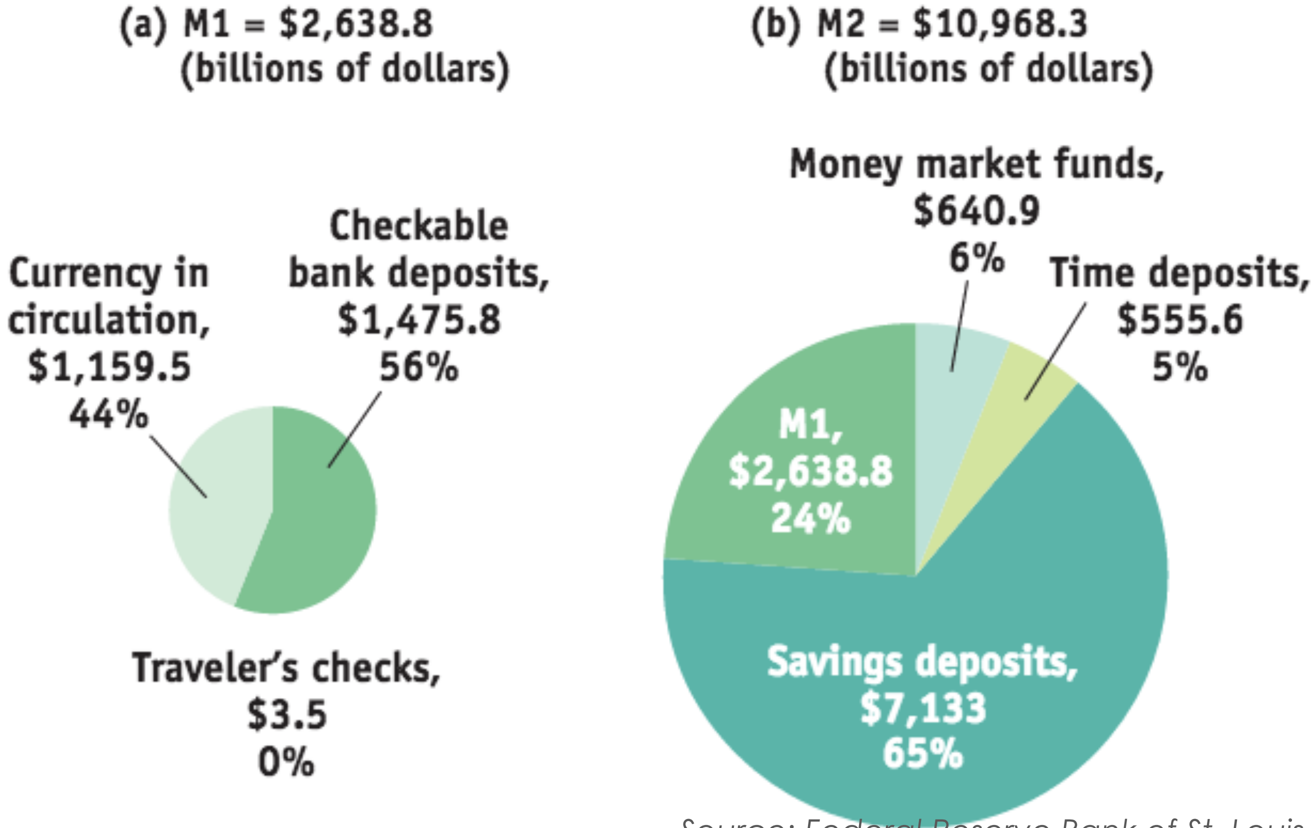
MONETARY AGGREGATES, 2013

M1 includes only the most liquid forms of money.

M2 includes ***near-moneys:*** financial assets that can't be directly used as a medium of exchange but can readily be converted into cash or checkable bank deposits.



Source: Federal Reserve Bank of St. Louis.

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LEARN BY DOING: PRACTICE QUESTION

Which of the following assets would you classify as being most liquid?

a) demand deposits

b) small-time deposits

c) a home

d) gold bullion

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THE MONETARY ROLE OF BANKS

Q: Only half of M1 is currency; where does the rest of the money supply come from?

A: Bank deposits make up half of M1 and most of M2.

How does the banking system create money?

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WHAT BANKS DO

Banks are financial intermediaries that use liquid assets (in the form of bank deposits) to finance the illiquid investments of borrowers.

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THE MONETARY ROLE OF BANKS

T-account: **a tool for analyzing a business's financial position by showing the business's assets and liabilities.**

Assets		Liabilities	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Bank reserves: **the currency that banks hold in their vaults plus their deposits at the Federal Reserve.**

The reserve ratio: **the fraction of bank deposits that a bank holds as reserves ($100,000/$1,000,000) = 10%.**

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HOW BANKS CREATE MONEY

The primary purpose of financial markets and institutions is to act as a bridge between lenders and borrowers.

By accepting deposits and making loans, banks and other financial institutions are also able to create money.

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HOW BANKS CREATE MONEY

Silas keeps a shoebox full of cash under his bed. Deciding to enter the twenty-first century, he deposits this cash at the bank. ***What's the effect of his $1,000?***

(a) Initial Effect Before Bank Makes a New Loan

Assets		Liabilities	
Loans	No change	Checkable deposits	+$1,000
Reserves	+$1,000		

(b) Effect When Bank Makes a New Loan

Assets		Liabilities
Loans	+$900	No change
Reserves	−$900	

Banks make their profit from loans, so eventually they will make new loans with 90% of Silas's money. (The rest they must hold in reserve.)

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HOW BANKS CREATE MONEY

The first bank lends $900 to Maya, who pays the money to Anne, who deposits it at her bank—and the cycle starts all over. Sound familiar?

	Currency in circulation	Checkable bank deposits	Money supply
First stage: Silas keeps his cash under his bed.	$1,000	$0	$1,000
Second stage: Silas deposits cash in First Street Bank, which lends out $900 to Maya, who then pays it to Anne Acme.	900	1,000	1,900
Third stage: Anne Acme deposits $900 in Second Street Bank, which lends out $810 to another borrower.	810	1,900	2,710

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RESERVES, BANK DEPOSITS, AND THE MONEY MULTIPLIER

Excess reserves: **a bank's reserves over and above its required reserves.**

We'll assume banks don't want to hold excess reserves.

Increase in bank deposits from $1,000 in excess reserves =

$$\$1{,}000 + [\$1{,}000 \times (1 - rr)] + [\$1{,}000 \times (1 - rr)^2] + [\$1{,}000 \times (1 - rr)^3] + \ldots$$

This can be simplified to $\$1{,}000/rr$.

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Assets		Liabilities	
Required reserves	$100	Deposits	$1,000
Loans	$400		
Treasury bills	$800		

Given this T-account and assuming the bank holds no excess reserves, what is the required reserve ratio?

a) 10%

b) 40%

c) 80%

d) 1%

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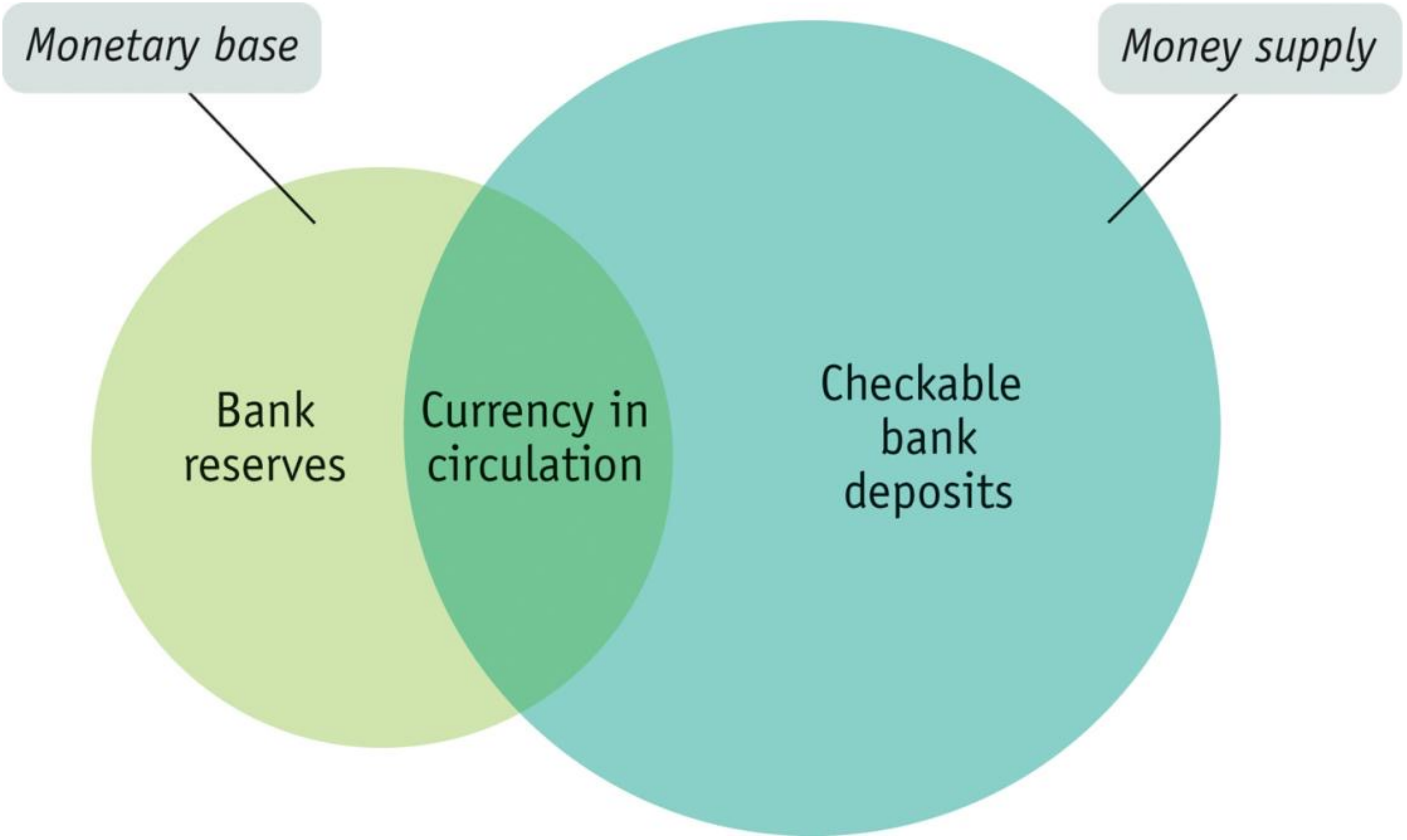
Suppose the required reserve ratio initially is 10% of bank deposits and is increased by the Fed to 20% of bank deposits. Holding everything else constant, this will:

a) reduce the size of the money multiplier.
b) cause the banking system to contract the level of bank deposits in the banking system.
c) change the value of the money multiplier from 10 to 5.
d) Answers (a), (b), and (c) are all correct.

THE MONEY MULTIPLIER IN REALITY

The ***monetary base*** is the sum of currency in circulation and bank reserves.



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THE MONEY MULTIPLIER IN REALITY

The determination of the money supply is more complicated than our simple model suggests:

It depends not only on the ratio of reserves to bank deposits but also on the fraction of the money supply that individuals choose to hold in the form of currency.

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THE MONEY MULTIPLIER IN REALITY

The *money multiplier* is the ratio of the money supply to the monetary base.

In normal times, the U.S. money multiplier for M1 is between 1.5 and 3.0.

Why isn't it 1/0.1 = 10?

People hold significant amounts of cash, which reduces bank deposits.

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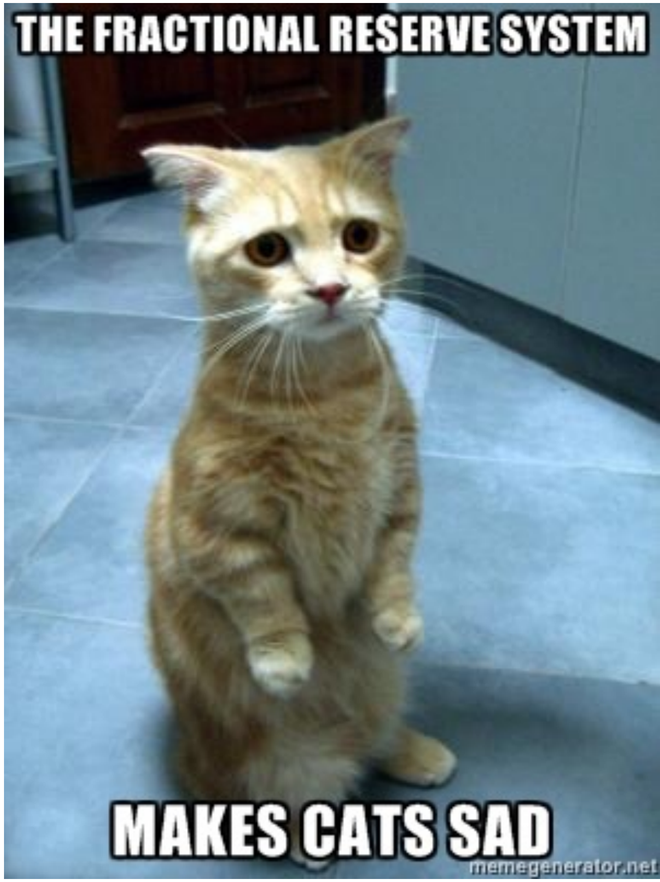
MONEY CREATION

Financial institutions operate as part of a *fractional reserve banking system.*

- **When you deposit money in a bank account, the bank is required to hold a part of it in its vault as cash**
  - ***(or else in an account with the regional Federal Reserve Bank).***



This system's leverage makes some nervous.

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THE PROBLEM OF BANK RUNS

Banks hold only a fraction of deposits on reserve. (The rest they use to make loans.)

> ***From time-to-time people get worried about the bank losing their money.***

Bank run: **a phenomenon in which many of a bank's depositors try to withdraw their funds because they fear a bank failure.**

Historically, runs have often proved contagious.

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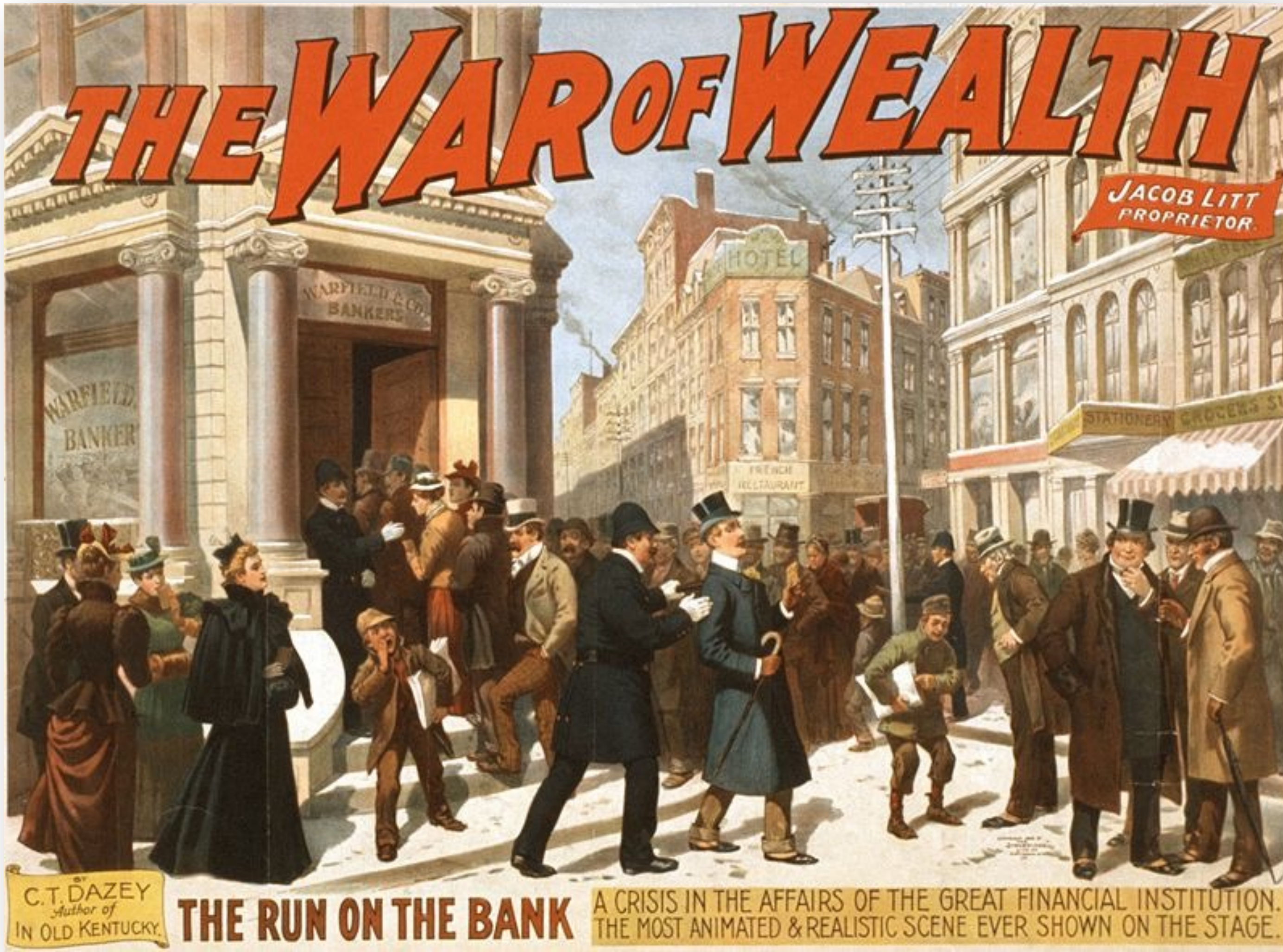
BANK PANICS

Bank run during the [Great Depression] in the United States, 1933.



A poster for the 1896 Broadway melodrama *The War of Wealth* depicts a typical 19th-century bank run in the United States.

BANK REGULATION

1. *Deposit insurance:* **a guarantee that a bank's depositors will be paid even if the bank can't come up with the funds. (The FDIC currently guarantees the first $250,000 of each account.)**

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BANK REGULATION

2. ***Capital requirements:*** **requirement that the owners of banks hold substantially more assets than the value of bank deposits.**

   Deposit insurance creates a well-known incentive problem: Banks can take more risks, since they are insured.

   To help motivate safe behavior, banks' capital is required to equal to 7% or more of their assets.

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BANK REGULATION

3. ***Reserve requirements:*** **rules set by the Federal Reserve that determine the minimum reserve ratio for a bank.**

   For example, in the United States, the minimum reserve ratio for checkable bank deposits is 10%.

4. ***The discount window:*** **an arrangement in which the Federal Reserve stands ready to lend money to banks in trouble.**

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ECONOMICS ▶ IN ACTION

IT'S A WONDERFUL BANKING SYSTEM

Bank runs still happen: Southern California's IndyMac: July 2008

Silicon Vally Bank: Mar. 2023

MULTIPLYING MONEY DOWN

If Silas can increase the money supply by pulling cash out from under his bed and injecting into the banking system, the reverse can happen anytime people carry cash.

Example: The Bank Runs of 1929-1933

	Currency in circulation	Checkable bank deposits	M1
	(billions of dollars)		
1929	$3.90	$22.74	$26.64
1933	5.09	14.82	19.91
Percent change	+31%	−35%	−25%

Source: U.S. Census Bureau (1975), *Historical Statistics of the United States.*

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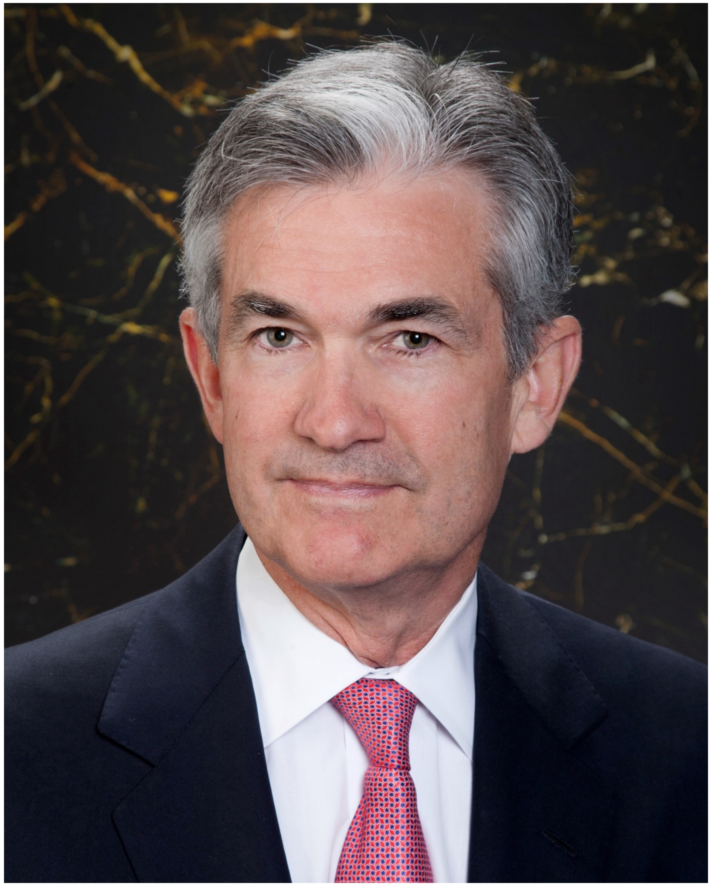
THE FEDERAL RESERVE SYSTEM

The ***Federal Reserve*** is a **central bank**; ***it oversees and regulates the banking system*** *and* ***controls the monetary base.***

Jerome Powell
16th chair of the Fed
Assumed office
Feb, 2018



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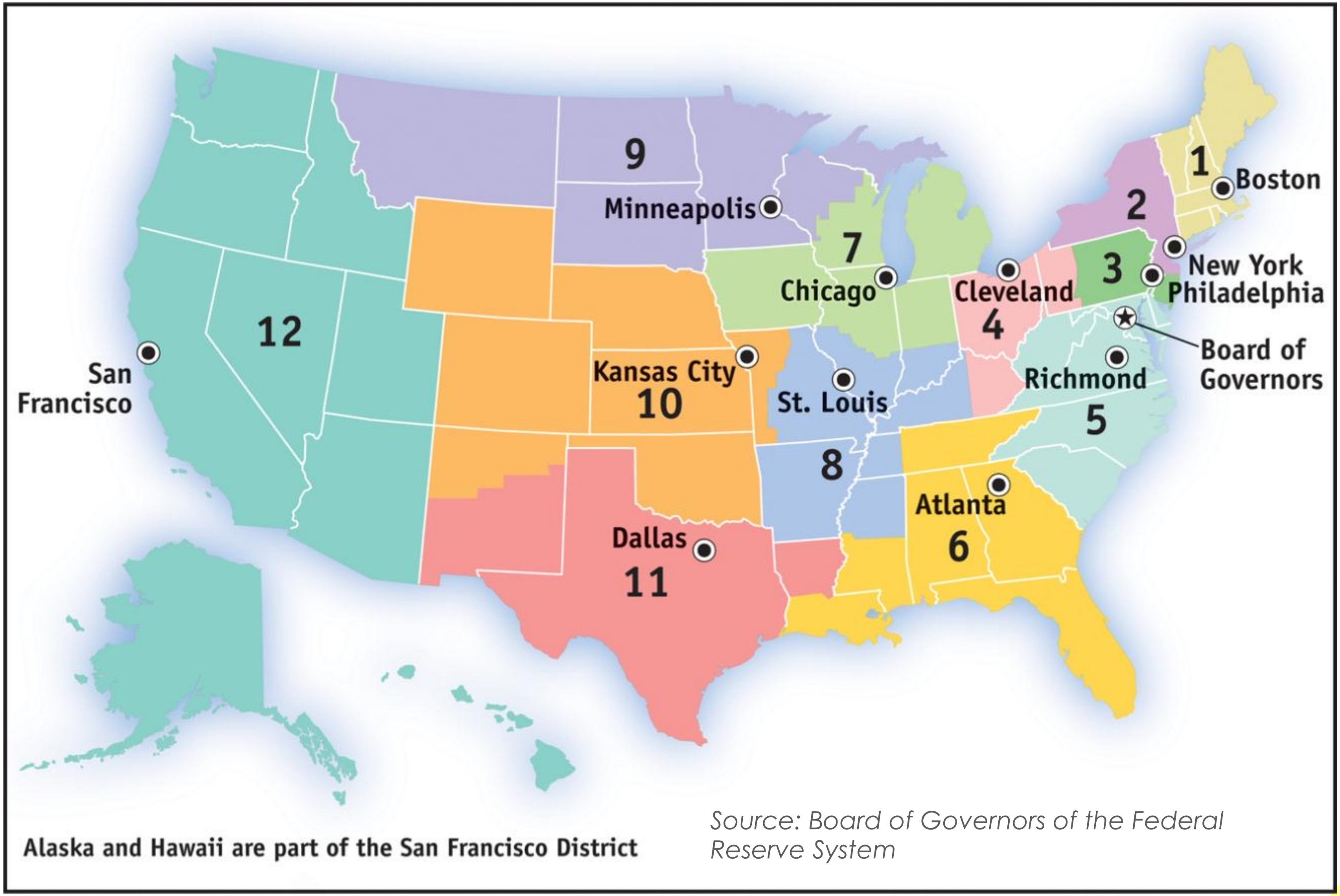
THE FEDERAL RESERVE SYSTEM



Source: Board of Governors of the Federal Reserve System

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RESERVE REQUIREMENTS AND THE DISCOUNT RATE

What does a bank do if it can’t meet the Fed’s reserve requirement?

The ***federal funds market*** allows banks that fall short of the reserve requirement to borrow funds from banks with excess reserves.

The ***federal funds rate*** is the interest rate determined in the federal funds market.

LIBOR (London interbank offered rate) is Europe’s equivalent to the federal funds rate.

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RESERVE REQUIREMENTS AND THE DISCOUNT RATE

The *discount rate* is the rate of interest the Fed charges on loans to banks.

Normally, the discount rate is set 1 percentage point above the federal funds rate in order to discourage banks from turning to the Fed when they are in need of reserves.

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COUPON BOND

10,000 10,000

THE UNITED STATES OF AMERICA

26925E

4¼%
TREASURY BOND
OF 1975-85
Dated April 5, 1960
Due May 15, 1985
Redeemable on and after
May 15, 1975

FOR VALUE RECEIVED

PROMISES TO PAY TO THE BEARER THE SUM OF

TEN THOUSAND DOLLARS

ON MAY 15, 1985, AND TO PAY INTEREST ON THE PRINCIPAL SUM FROM THE DATE HEREOF, AT THE RATE OF FOUR AND ONE-QUARTER PERCENT PER ANNUM, PAYABLE ON A SEMIANNUAL BASIS ON NOVEMBER 15, 1960, AND THEREAFTER ON MAY 15 AND NOVEMBER 15 IN EACH YEAR UNTIL THE PRINCIPAL HEREOF SHALL BE PAYABLE, UPON PRESENTATION AND SURRENDER OF THE INTEREST COUPONS HERETO ATTACHED AS THEY SEVERALLY MATURE, AT THE TREASURY DEPARTMENT, WASHINGTON, D. C., OR, AT THE HOLDER'S OPTION, AT ANY AGENCY OR AGENCIES IN THE UNITED STATES WHICH THE SECRETARY OF THE TREASURY MAY FROM TIME TO TIME DESIGNATE FOR THE PURPOSE. THIS BOND IS ONE OF A SERIES OF BONDS OF THE UNITED STATES, AUTHORIZED BY THE SECOND LIBERTY BOND ACT, AS AMENDED, ISSUED PURSUANT TO AND SUBJECT TO THE TERMS AND CONDITIONS OF TREASURY DEPARTMENT CIRCULAR NO. 1040, DATED APRIL 4, 1960, AND DESIGNATED 4¼ PERCENT TREASURY BONDS OF 1975-85. ALL OR ANY OF THE BONDS OF SAID SERIES MAY BE REDEEMED, AT THE OPTION OF THE UNITED STATES, ON AND AFTER MAY 15, 1975, AT PAR AND ACCRUED INTEREST, ON ANY INTEREST DAY OR DAYS, ON FOUR MONTHS' NOTICE OF REDEMPTION GIVEN IN SUCH MANNER AS THE SECRETARY OF THE TREASURY SHALL PRESCRIBE. IN CASE OF PARTIAL REDEMPTION THE BONDS TO BE REDEEMED WILL BE DETERMINED BY SUCH METHOD AS MAY BE PRESCRIBED BY THE SECRETARY OF THE TREASURY. FROM THE DATE OF REDEMPTION DESIGNATED IN ANY SUCH NOTICE, INTEREST ON THE BONDS CALLED FOR REDEMPTION SHALL CEASE. THE INCOME DERIVED FROM THIS BOND IS SUBJECT TO ALL TAXES IMPOSED UNDER THE INTERNAL REVENUE CODE OF 1954. THIS BOND IS SUBJECT TO ESTATE, INHERITANCE, GIFT OR OTHER EXCISE TAXES, WHETHER FEDERAL OR STATE, BUT IS EXEMPT FROM ALL TAXATION NOW OR HEREAFTER IMPOSED ON THE PRINCIPAL OR INTEREST HEREOF BY ANY STATE, OR ANY OF THE POSSESSIONS OF THE UNITED STATES, OR BY ANY LOCAL TAXING AUTHORITY. THIS BOND, UPON THE DEATH OF THE OWNER, WILL BE REDEEMED AT THE OPTION OF THE DULY CONSTITUTED REPRESENTATIVES OF THE DECEASED OWNER'S ESTATE, AT PAR AND ACCRUED INTEREST, IF IT CONSTITUTES PART OF SUCH ESTATE AND THE PROCEEDS ARE TO BE APPLIED TO THE PAYMENT OF

OPEN-MARKET OPERATIONS

When the Fed buys anything (even apples), reserves increase.

Because government bonds can be stored indefinitely and are easy to buy and sell on the open market, the Fed chooses this commodity to buy and sell daily.

> *The Fed can buy and sell billions of dollars' worth of government bonds in minutes.*

The Fed usually buys and sells short-term bonds called Treasury bills, or T-bills

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OPEN-MARKET OPERATIONS

The Fed does not buy bonds directly from the government.

When a central bank buys government debt directly from the government, it is lending directly to the government—in effect printing money to finance the government's budget deficit.

The Fed's decisions are independent of the government.

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LEARN BY DOING: DISCUSS

With a partner, discuss: If a president could pressure the chairman of the Federal Reserve to do his bidding, what would happen?

Do you think inflation be higher or lower if the Fed were not independent? Why?

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OPEN-MARKET OPERATIONS

What it looks like:

Open-market operations are the principal tool of monetary policy: the Fed can increase or reduce the monetary base by buying or selling government debt to banks.

The Federal Reserve's Assets and Liabilities:

Assets	Liabilities
Government debt (Treasury bills)	Monetary base (currency in circulation + bank reserves)

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OPEN-MARKET OPERATIONS

To *increase* the money supply, the Fed must inject reserves into the system.

$

To *decrease* the money supply, the Fed must remove reserves from the system.

$

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OPEN-MARKET OPERATIONS

An open-market purchase of $100 million: banks gain reserves.

	Assets		Liabilities	
Federal Reserve	Treasury bills	+$100 million	Monetary base	+$100 million

	Assets		Liabilities	
Commercial banks	Treasury bills	−$100 million	No change	
	Reserves	+$100 million		

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OPEN-MARKET OPERATIONS

An open-market sale of $100 million: bank reserves decline

	Assets		Liabilities	
Federal Reserve	Treasury bills	−\$100 million	Monetary base	−\$100 million

	Assets		Liabilities
Commercial banks	Treasury bills	+\$100 million	No change
	Reserves	−\$100 million	

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SHOW ME THE MONEY

Q: Where does the Fed gets the funds to purchase U.S. Treasury bills from banks?

A: It simply creates them with a stroke of the pen (a click of the mouse) that credits the banks' accounts with extra reserves.

(The Fed does NOT need to print money.)

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HOW OPEN-MARKET OPERATIONS WORK

If the Fed wants to increase the money supply, it will buy T-bills:

To pay for the T-bills → Fed electronically ↑reserves of the seller → With more reserves, bank ↑ loans → M↑ as the money creation process ripples through the economy

If the Fed wants to decrease the money supply, it will sell T-bills:

Fed sells T-bills → ↓reserves of the buyer → With fewer reserves, bank ↓ loans → M↓ as the money creation process ripples in *reverse* through the economy

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LEARN BY DOING: PRACTICE QUESTION

Assets		Liabilities	
Required reserves	$100	Deposits	$1,000
Loans	$400		
Treasury bills	$800		

Suppose this is the only bank and all money is held in this bank. There are no excess reserves. If the Fed makes an open-market sale of $50 worth of T-bills to this bank, what will happen to the money supply after all adjustments are made?

a) **The money supply will increase by $50.**

b) **The money supply will decrease by $50.**

c) **The money supply will increase by $500.**

d) **The money supply will decrease by $500.**

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How the Federal Reserve implements monetary policy, moving the interest rate to affect aggregate output?

THE DEMAND FOR MONEY

There is a price to be paid for the convenience of holding money.

The Opportunity Cost of Holding Money

We all carry some cash around for the convenience.

When we do, we give up interest income we'd collect if that spending power were in an interest-bearing asset like a bond.

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INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Individuals and firms trade off the benefit of holding cash versus the benefit of holding interest-bearing nonmonetary assets.

Selected interest rates, 2007 (before the financial crisis)

One-month certificates of deposit (CDs)	5.30%
Interest-bearing demand deposits	2.30%
Currency	0

Source: Federal Reserve Bank of St. Louis

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INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Short-term interest rates: **the interest rates on financial assets that mature within six months or less.**

Long-term interest rates: **interest rates on financial assets that mature a number of years in the future.**

The higher the interest rate, the higher the opportunity cost of holding money.

The lower the interest rate, the lower the opportunity cost of holding money.

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INTEREST RATES AND THE OPPORTUNITY COST OF HOLDING MONEY

Although there are many different interest rates and they don't move exactly together, as a rule most short-term rates tend to move in the same direction.

	June 2007	June 2008
Federal funds rate	5.25%	2.00%
One-month certificates of deposit (CDs)	5.30%	2.50%
Interest-bearing demand deposits	2.30%	1.24%
Currency	0	0
CDs minus interest-bearing demand deposits (percentage points)	**3.00**	**1.26**
CDs minus currency (percentage points)	**5.30**	**2.50**

Source: Federal Reserve Bank of St. Louis

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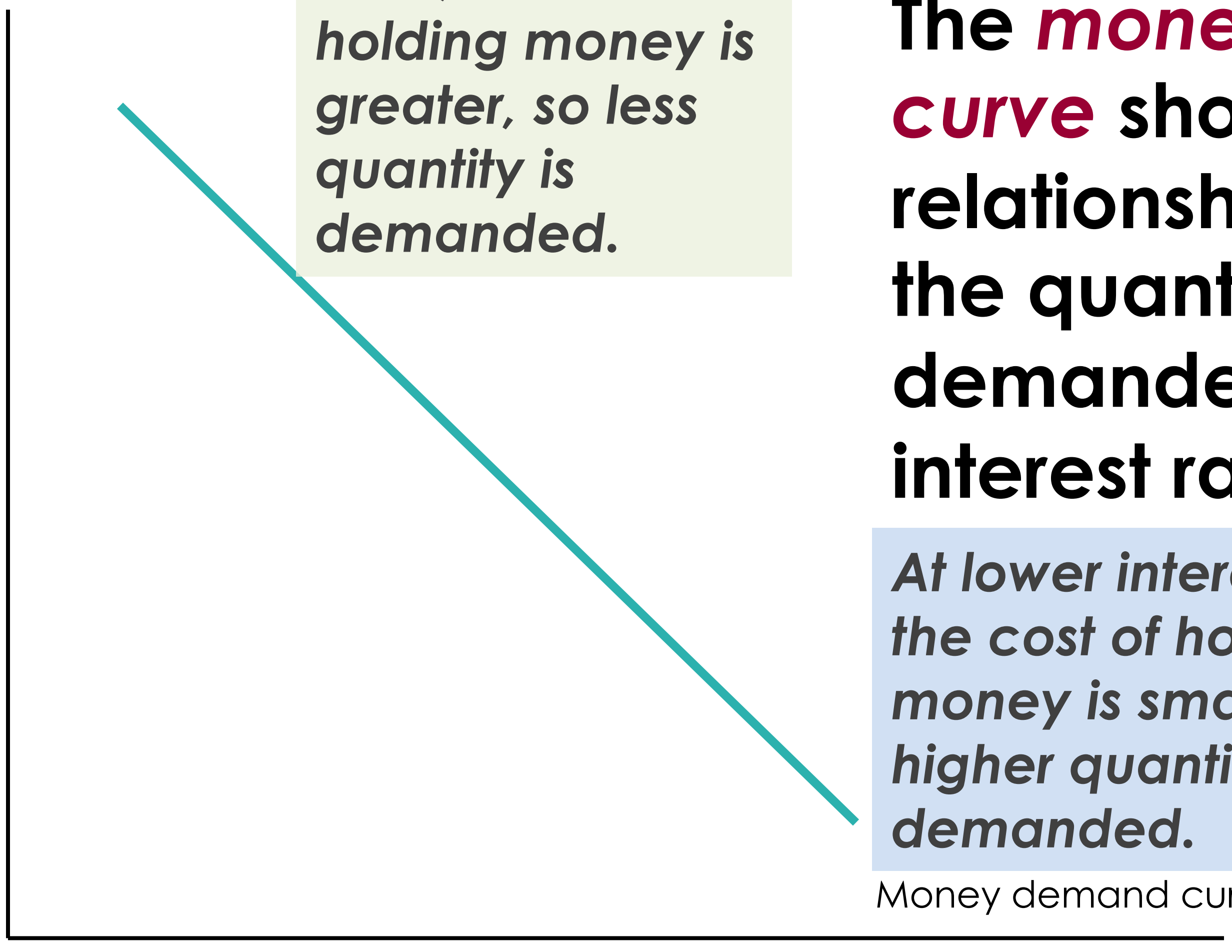
THE MONEY DEMAND CURVE



At higher interest rates, the cost of holding money is greater, so less quantity is demanded.

The *money demand curve* shows the relationship between the quantity of money demanded and the interest rate.

At lower interest rates, the cost of holding money is smaller, so a higher quantity is demanded.

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SHIFTS OF THE REAL MONEY DEMAND CURVE

The following factors shift the money demand curve:

- **Changes in aggregate price level**
  - *Higher prices mean we need more money for transactions (and vice versa).*
- **Changes in real GDP**
  - *More goods and services produced and sold means we need more money (and vice versa).*
- **Changes in technology**
  - *The ease of credit card payments reduces the need for cash.*
- **Changes in institutions**
  - *After 1980 banks were allowed to offer interest on checking accounts. This decreased the cost of holding money, and money demand increased.*

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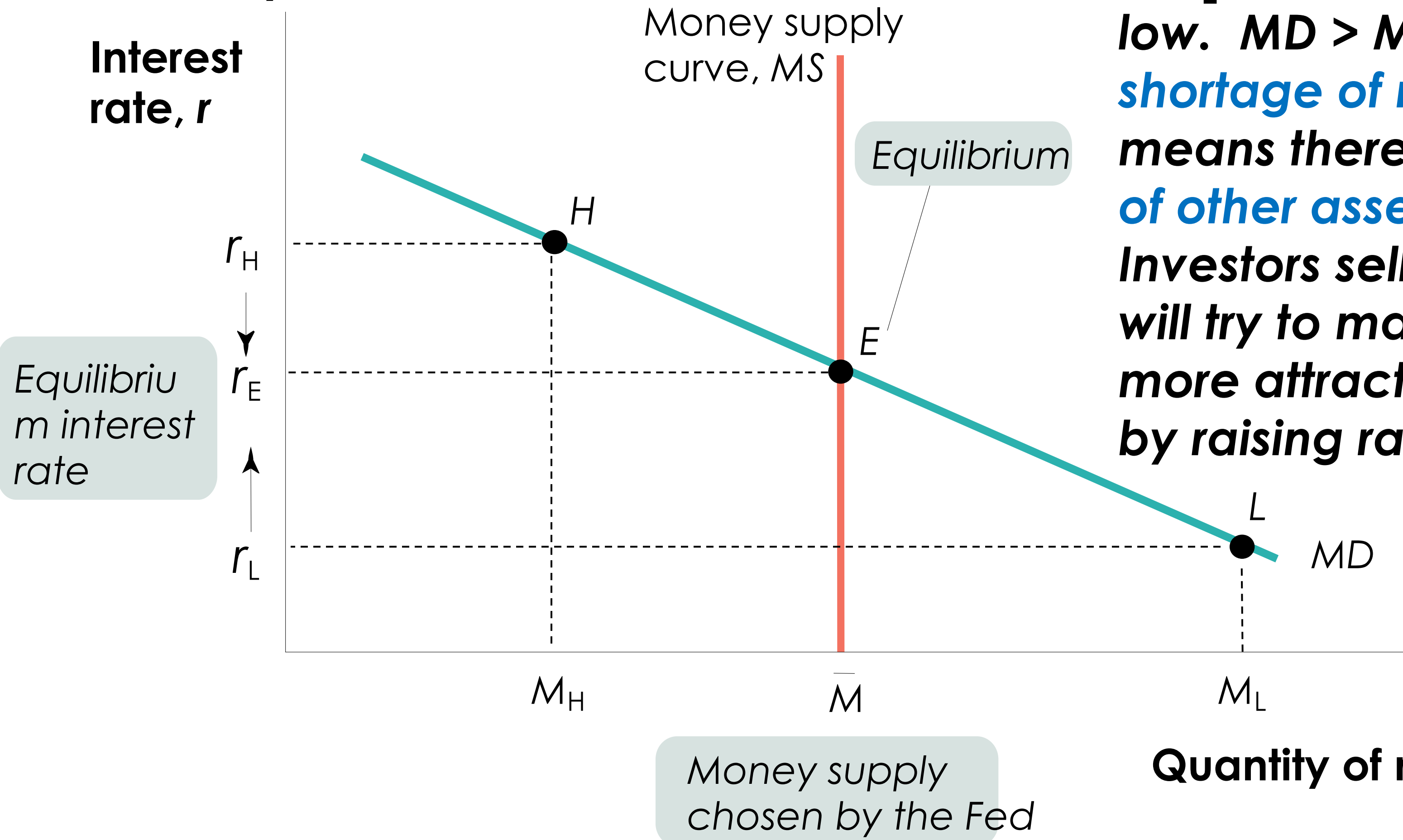
EQUILIBRIUM IN THE MONEY MARKET

At r_H, interest rates are high. MS > MD, and a surplus of money means there is a shortage of other assets like bonds. Investors selling bonds will realize they can reduce interest rates and still find buyers.



At r_L, interest rates are low. MD > MS, and a shortage of money means there is a surplus of other assets like bonds. Investors selling bonds will try to make them more attractive to buyers by raising rates.

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MONETARY POLICY AND THE INTEREST RATE

By adjusting the money supply up or down, the Fed can set the interest rate.

The Fed sets a **target federal funds rate and undertakes the appropriate open-market operations.**

- ***Target federal funds rate:*** the Federal Reserve's desired federal funds rate.

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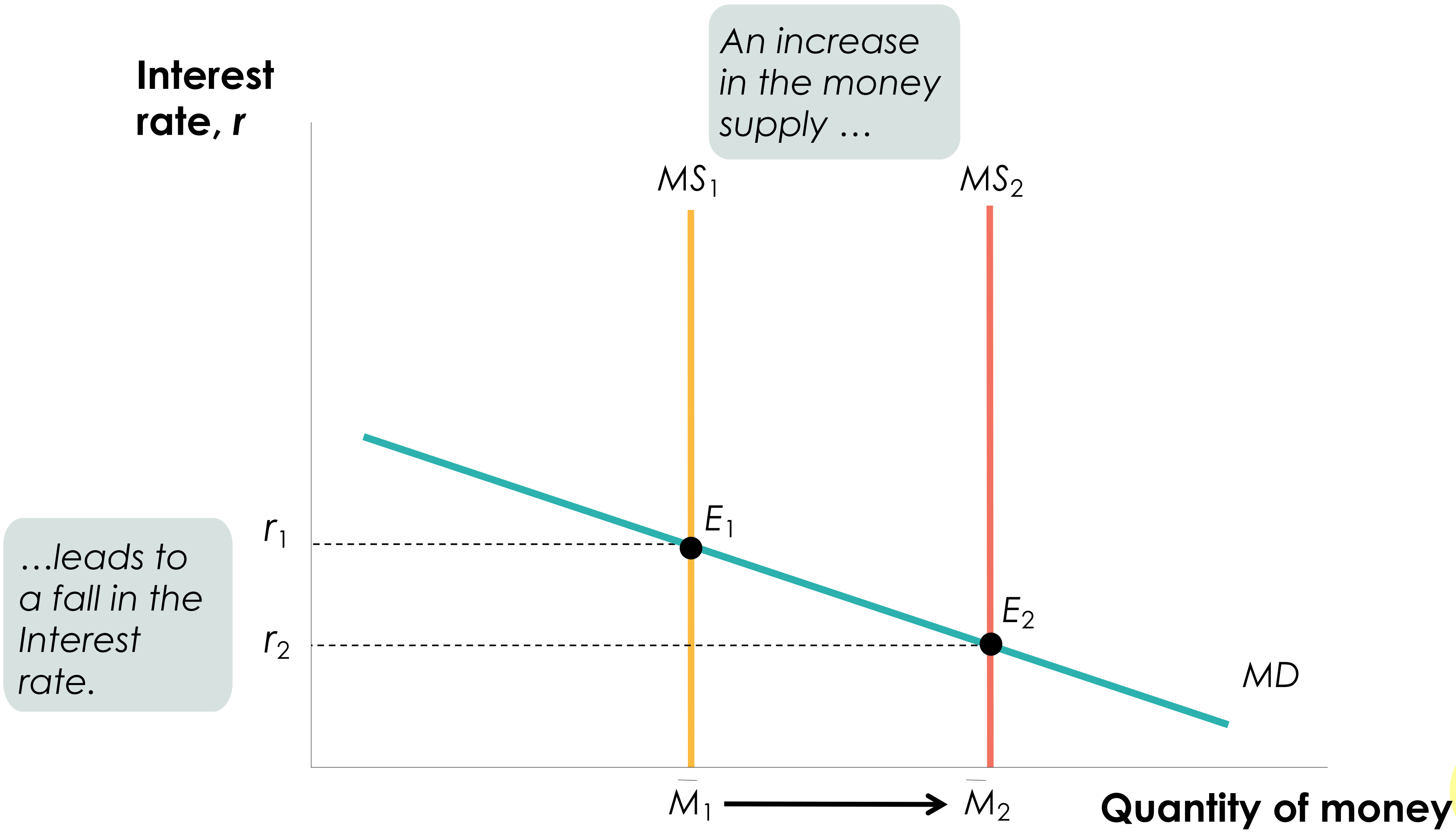
MONETARY POLICY AND THE INTEREST RATE

The Effect of an Increase in the Money Supply on the Interest Rate



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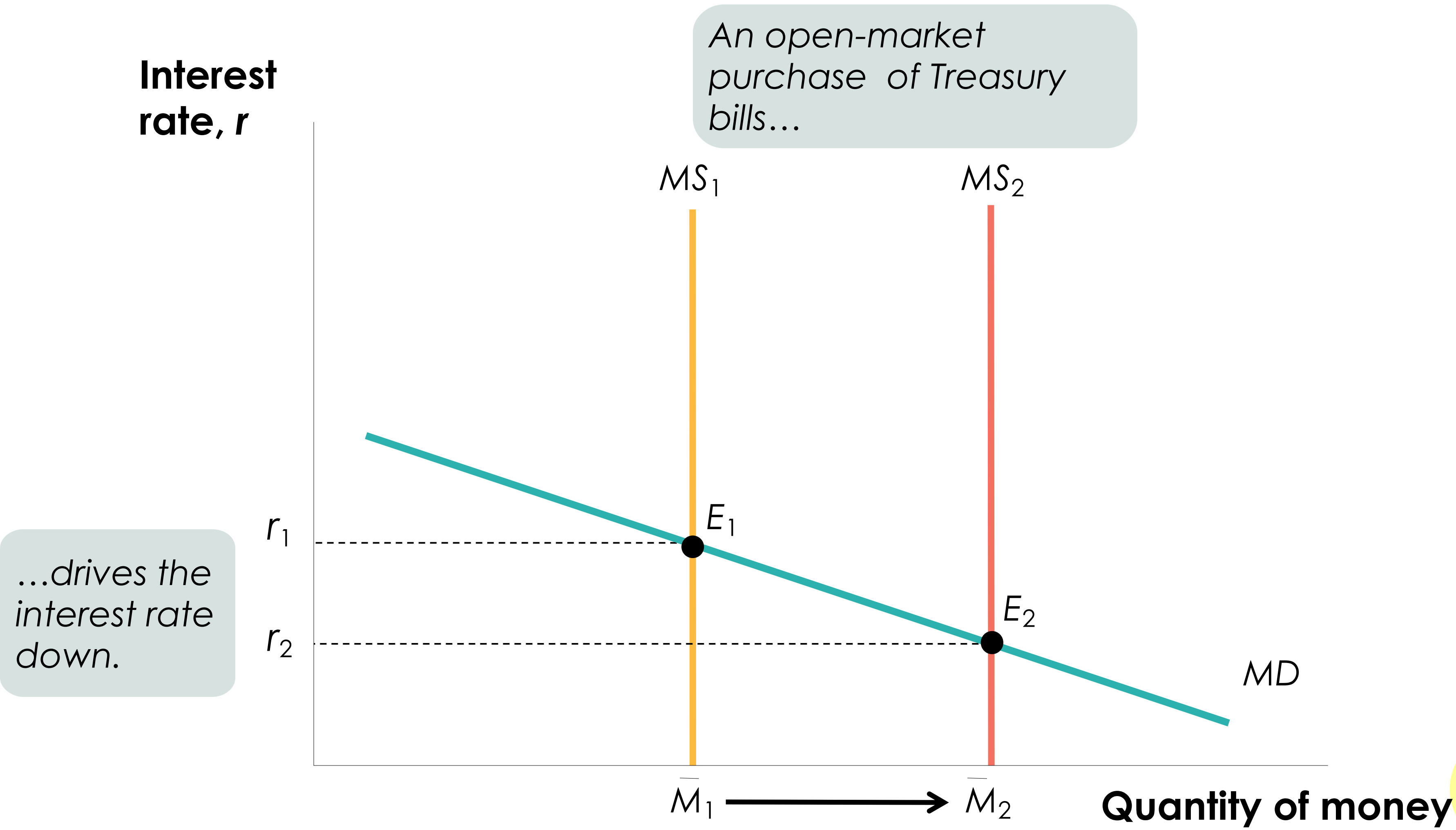
SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to push the interest rate down to the target rate.



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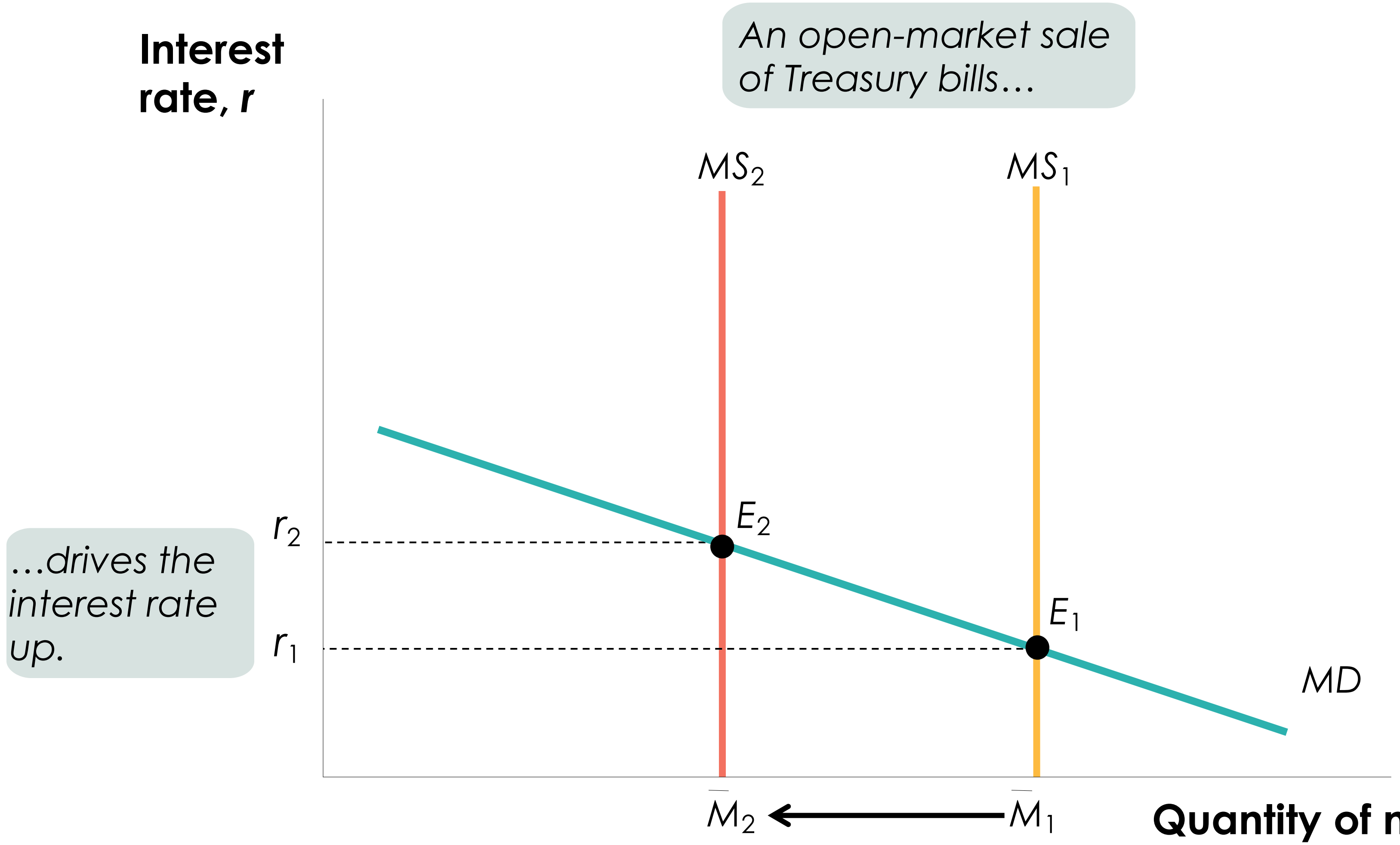
SETTING THE FEDERAL FUNDS RATE

The Fed uses open-market operations to pull the interest rate up to the target rate.



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MONETARY POLICY AND AGGREGATE DEMAND

Expansionary monetary policy: **monetary policy that increases aggregate demand (also called "easy money policy").**

Contractionary monetary policy: **monetary policy that reduces aggregate demand (also called "tight money policy").**

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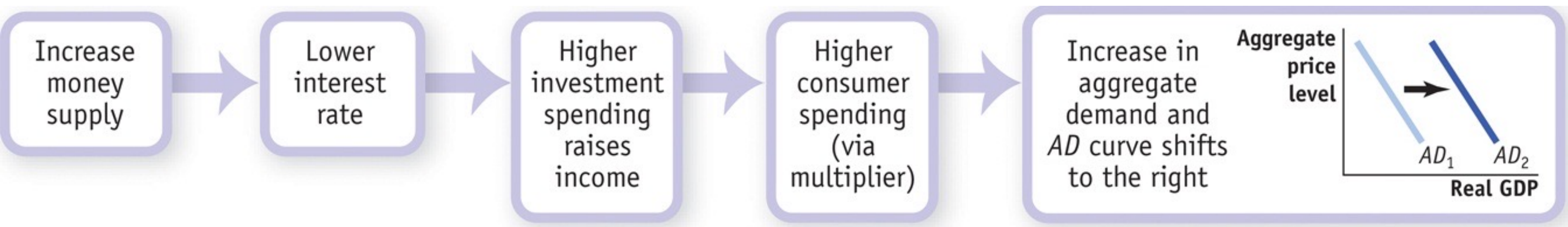
MONETARY POLICY AND AGGREGATE DEMAND

How the expansionary monetary policy cycle works:



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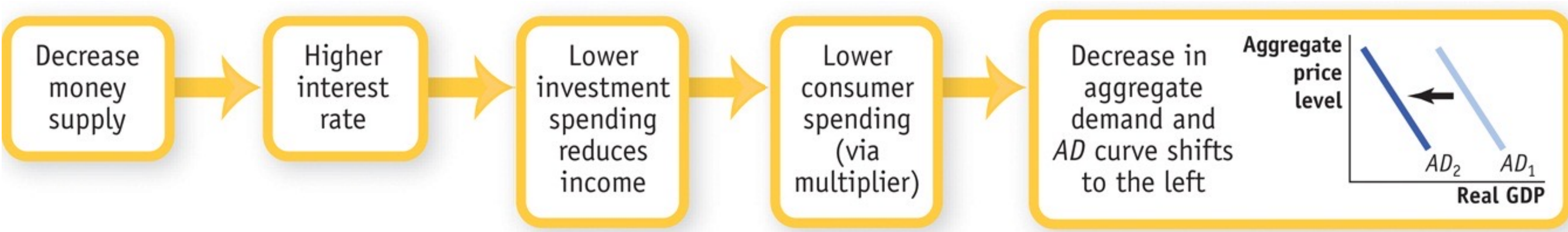
MONETARY POLICY AND AGGREGATE DEMAND

How the contractionary monetary policy cycle works:



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MONETARY POLICY IN PRACTICE

How does the Fed decide whether to use expansionary or contractionary monetary policy?

And how does it decide how much is enough?

Policy makers try to fight recessions, as well as to ensure *price stability*: low (though usually not zero) inflation.

Actual monetary policy reflects a combination of these goals.

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THE EUROPEAN CENTRAL BANK

Created in January 1999

Located in Frankfurt, Germany

Represents a (combined) economy as large as that of the United States.

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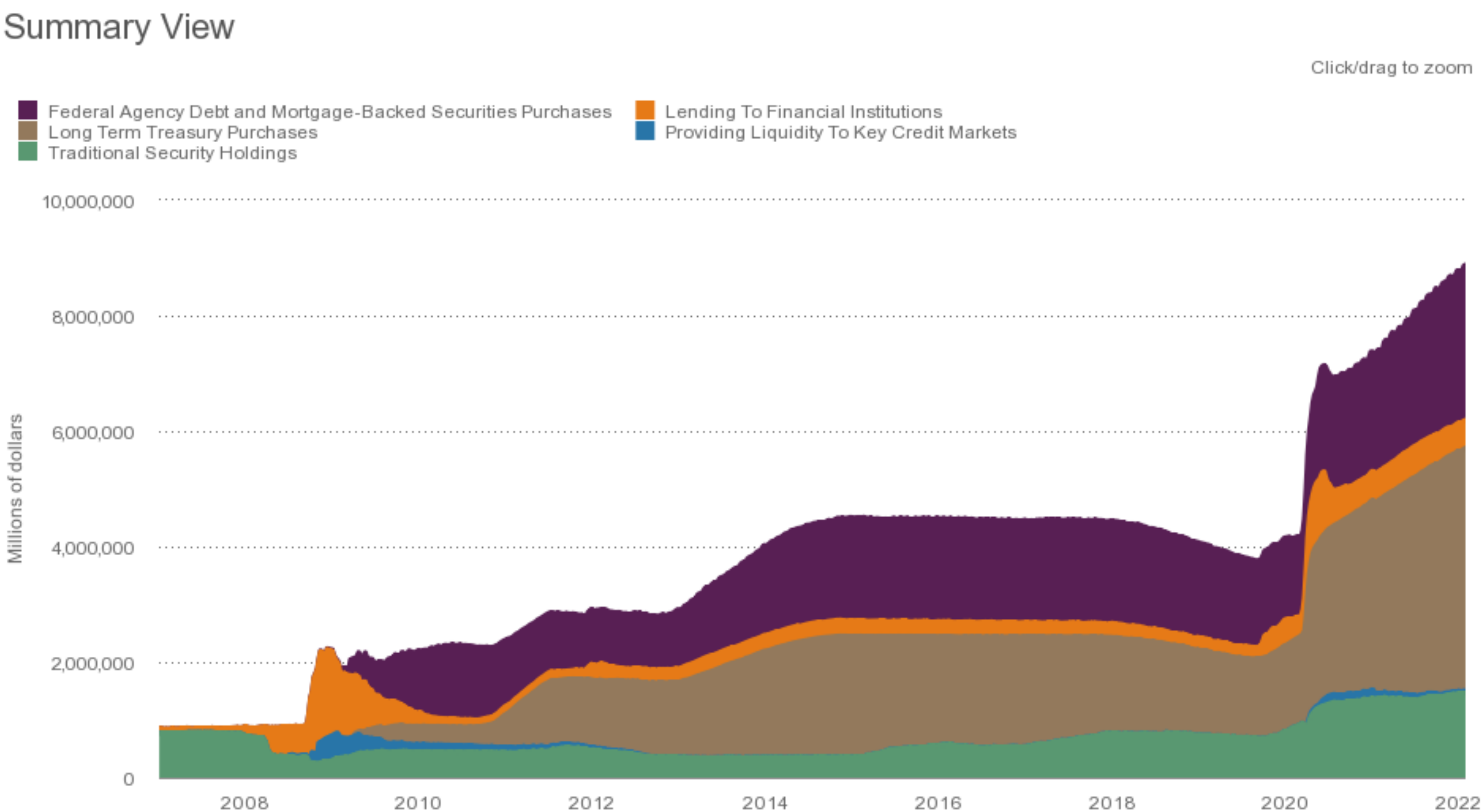
ECONOMICS IN ACTION

THE FED'S BALANCE SHEET



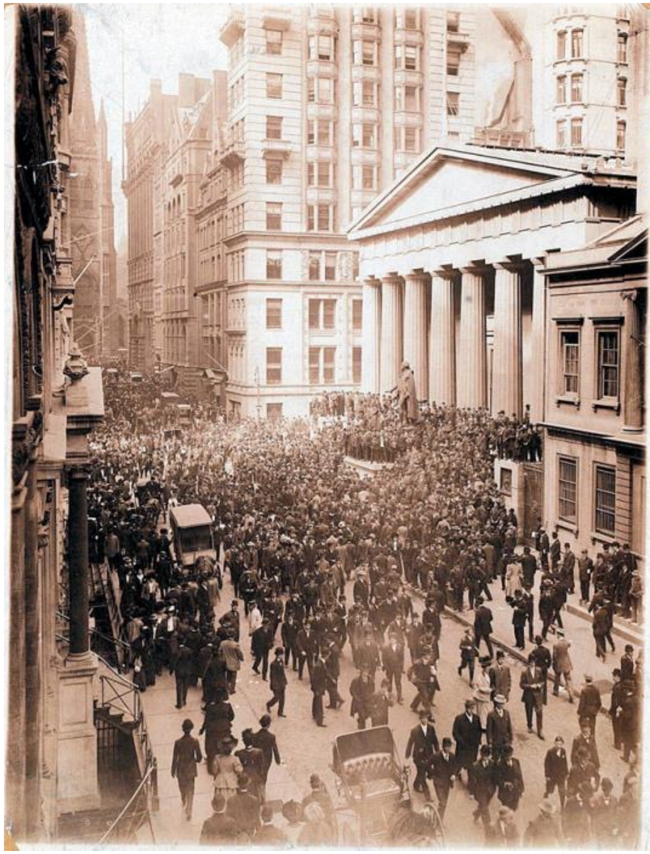
Source: Federal Reserve Bank of Cleveland calculations based on data from Federal Reserve Board and Haver Analytics.

THE EVOLUTION OF THE AMERICAN BANKING SYSTEM

The crisis in American banking in the early twentieth century

The main problem afflicting the system was that the money supply was not sufficiently responsive: It was difficult to shift currency around the country to respond quickly to local economic changes.

Bank panics (bank runs) were too common.



The Panic of 1907: a foreshadowing of the financial crisis of 2008? Large losses from risky speculation destabilized the banking system.

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THE EVOLUTION OF THE AMERICAN BANKING SYSTEM

Responding to banking crises: the creation of the Federal Reserve

The panic of 1907 convinced many that the time for central control of bank reserves had come.

In 1913 the Federal Reserve was created.

Now bank reserves were centralized and standardized—but not immune to bank runs.

A series of bank runs in 1930, 1931, and 1933 convinced the Fed that **banks needed more regulation.**

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THE EVOLUTION OF THE AMERICAN BANKING SYSTEM

In **1932**, the **Reconstruction Finance Corporation (RFC)** was established and given the authority to make loans to banks in order to stabilize the banking sector.

Also in **1932**, the **Glass-Steagall Act** created federal deposit insurance and increased the ability of banks to borrow from the Federal Reserve system.

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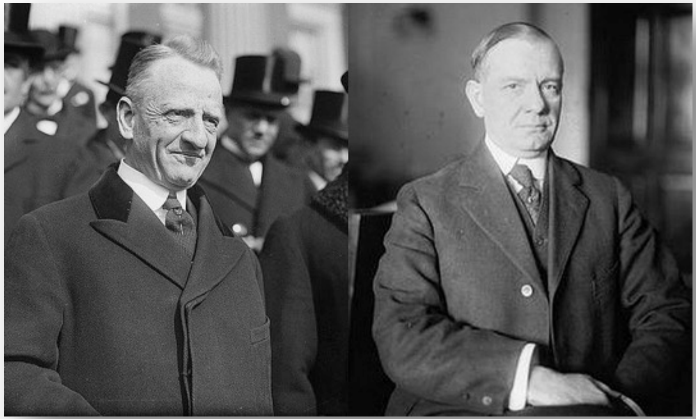
THE EVOLUTION OF THE AMERICAN BANKING SYSTEM

Another **Glass-Steagall Act** **(1933)** separated banks into two categories:

Commercial banks: depository banks that accepted deposits and were covered by deposit insurance.

Investment banks: banks which engaged in creating and trading financial assets (stocks and corporate bonds) but were not covered by deposit insurance because their activities were considered riskier.

Public acceptance of deposit insurance finally stopped the bank runs of the Great Depression.



Sen. Carter Glass (D-Va.) and Rep. Henry B. Steagall (D-Ala.)

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THE SAVINGS AND LOAN CRISIS OF THE 1980S

The savings and loan (thrift) crisis of the 1980s arose because Congress deregulated S&Ls, allowing them to engage in risky speculation.

The intent was to make them to offer higher rates of return, like banks, in an era of high inflation.

They incurred huge losses.

Depositors in failed S&Ls were compensated with taxpayer funds because they were covered by deposit insurance.

The crisis caused steep losses in the financial and real estate sectors, resulting in a recession in the early 1990s.

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THE FINANCIAL CRISIS OF 2008

During the mid-1990s, the hedge fund LTCM (Long-Term Capital Management) used huge amounts of leverage to speculate in global financial markets, incurred massive losses, and collapsed.

> ***Leverage:*** financing an investment with **borrowed funds.**

LTCM was so large that it caused balance sheet effects for firms globally, leading to the prospect of a vicious circle of deleveraging.

> ***Balance sheet effect:*** the reduction in a firm's net worth due to falling asset prices.
>
> ***Vicious circle of deleveraging:*** asset sales to cover losses forcing creditors to call in their loans, forcing sales of more assets and causing further declines in asset prices.

As a result, credit markets around the world froze. The New York Fed coordinated a private bailout of LTCM and revived world credit markets.

LTCM
The financial technology company

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THE FINANCIAL CRISIS OF 2008

2008—The worldwide financial system was in crisis, and banks and other financial institutions wanted to borrow more than $2 trillion. What could be done?

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THE FINANCIAL CRISIS OF 2008

Subprime lending during the U.S. housing bubble of the mid-2000s spread through the financial system via securitization.

> ***Subprime lending:*** lending to home buyers who don't meet the usual criteria for being able to make the payments.
>
> ***Securitization:*** a pool of loans assembled and shares of it sold to investors.

When the bubble burst, massive losses by banks and other financial institutions led to widespread collapse in the financial system.

To prevent another Great Depression, the Fed and the U.S. Treasury expanded lending to banks and other financial institutions, provided capital through the purchase of bank shares, and purchased private debt.

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LEARN BY DOING: APPLICATION VIDEO

Many Americans still don't understand what happened to the economy in 2008. How did it all go so bad so quickly? Who is responsible? How effective has the response from Washington and Wall Street been? Those are the questions at the heart of *FRONTLINE's* "Inside the Meltdown." Click here. (56:23 minutes)

http://www.pbs.org/wgbh/pages/frontline/meltdown/view/

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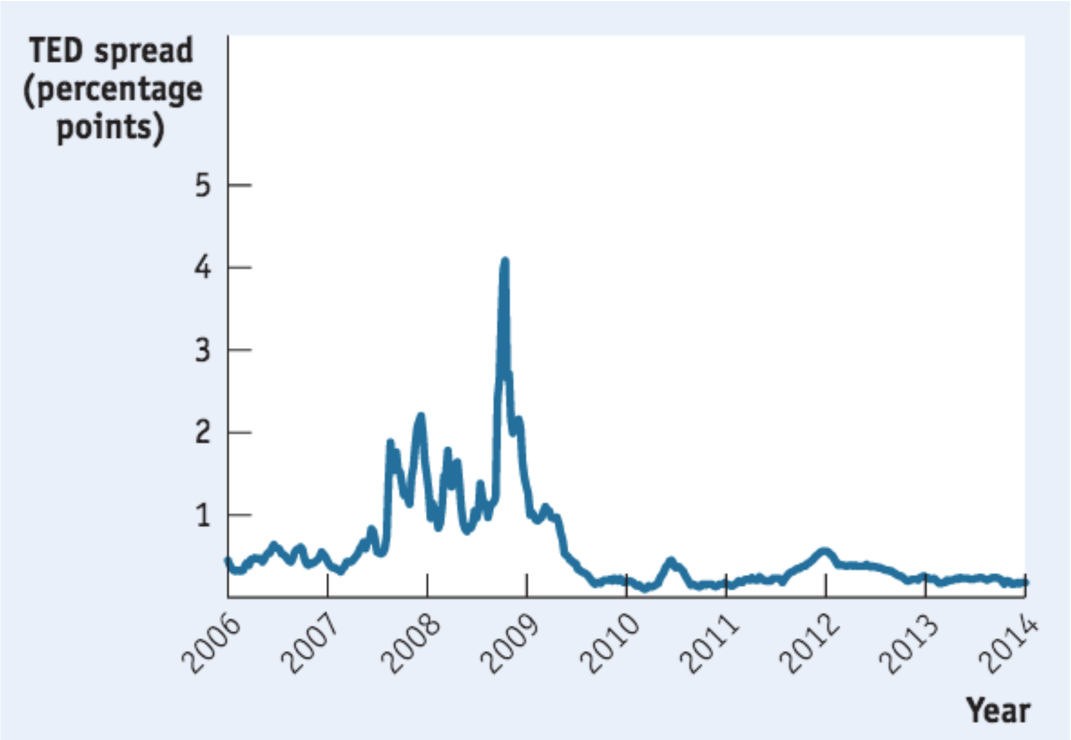
THE FINANCIAL CRISIS OF 2008

Because much of the crisis originated in nontraditional bank institutions, the crisis of 2008 indicated that a wider safety net and broader regulation are needed in the financial sector.



The TED spread: widely used as a measure of financial stress and perceived risk. (It's the difference between the interest rate at which banks lend to each other and the interest rate on U.S. government debt.)

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ECONOMICS ▶ IN ACTION

REGULATION AFTER THE 2008 CRISIS

In July 2010, President Obama signed the Dodd-Frank Act (Wall Street Reform and Consumer Protection Act).

- Gives a special government committee, the Financial Stability Oversight Council, the right to designate institutions as "systemically important" and thus subject to normal bank regulation (even though they are not traditional banks).
- Established new rules on the trading of derivatives: They must now be traded in exchanges so their prices and volume are more transparent.

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Quantitative Easing

- Quantitative easing (QE) is an unconventional monetary policy tool used by central banks to stimulate the economy when traditional monetary policy tools, such as lowering interest rates, become ineffective or reach their lower bounds.

- Under quantitative easing, a central bank creates new money electronically and uses it to purchase financial assets, such as government bonds and other securities, from banks and other financial institutions.

- By purchasing these assets, the central bank increases the demand for them, raising their prices and effectively lowering their yields (interest rates).

- Lower long-term interest rates make borrowing cheaper for businesses and households, which in turn promotes investment, spending, and economic growth.

- While QE can be an effective tool to stimulate economic growth, it also carries risks, such as the potential for higher inflation, asset price bubbles, and the distortion of financial markets.

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LEARN BY DOING: PRACTICE QUESTION

The Federal Funds rate is the rate that banks are charged when they borrow from the Fed.

a) True

b) False

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LEARN BY DOING: BUSINESS CASE

THE PERFECT GIFT: CASH OR A GIFT CARD?

The best-selling single item for more than 80% of top U.S. retailers? Gift cards.

Retailers know that breakage rates are high (unuse)- more than $1billion/year

Recipients will often sell these cards at a discount to get cash.

hoto Works

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